

Design of Mindful Eating Interactive System Based on Mixed Reality and Multimodal Generative Art

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Abstract: To address low long-term adherence in traditional mindful eating therapy, caused by insufficient personalization and limited sensory engagement, we propose an innovative system that combines Mixed Reality (MR) with multi-modal generative digital art. This framework enhances mindful eating by offering a personalized, immersive, and artistic experience. The system integrates MR glasses and a mobile app with the Gourmet Flow dual-end system, blending virtual interaction and AI to create a co-creative, sensory-rich dining experience. The process begins with the mobile camera analyzing food data, which is transmitted to the MR terminal to generate an adaptive virtual dining environment. Users can choose their virtual dining setting while interacting with the environment. Additionally, interactive music visualization and generative synthesis, triggered by olfactory and taste feedback, produce unique color and shape combinations, forming audio-visual art. This approach enhances both physical and mental satisfaction, reshaping mindful eating through artistic expression and offering insights into MR and multi-modal generative art in human-computer co-creation.

Keyword: Mixed reality, Multimodal Generative Art, Gourmet Flow System, TouchDesigner platform

1 Introduction

In recent years, mindful eating has gained attention as a therapeutic approach to improve eating habits and overall well-being[1-2]. However, traditional practices face challenges such as low long-term adherence, limited personalization, and insufficient sensory engagement, which hinder their effectiveness[3]. With advancements in mixed reality (MR) and generative AI, new opportunities have arisen to enhance mindful eating through personalized, immersive, and multi-sensory experiences[4]. These technologies enable the creation of virtual environments, AI-driven dynamic content, and real-time feedback, addressing the limitations of conventional methods[5]. This paper explores how MR and generative AI can be used to design interactive mindful eating experiences, fostering deeper engagement, better adherence, and improved

health outcomes by incorporating sensory feedback, creative interaction, and reflection[6].

This paper examines how cutting-edge technologies, such as MR and Multimodal Generative Art generative AI, can be utilized to design interactive digital mindful eating experiences. The proposed framework aims to enhance sensory engagement, personalization, and long-term adherence, ultimately improving the effectiveness of mindful eating. The system not only promotes mindful eating but also encourages users to engage with the food experience in creative, artistic, and reflective ways [7]. By combining virtual environments, sensory feedback, and dynamic art content, this design will deepen users' mindful eating experiences, improve adherence, and contribute to better health and well-being [8].

In the following chapters, we will explore the theoretical foundations of mindful eating, the role of digital technologies in enhancing these practices, and the principles for designing interactive digital mindful eating experiences. This paper aims to provide fresh insights and practical recommendations for the field of digital health interventions, particularly within mindfulness research, and demonstrate the potential of MR and AI technologies in transforming food and eating habits.

2 Methods

2.1 Research Objectives

By gamifying and multi-sensorizing the eating experience, It also enhances interaction. we can significantly enhance its enjoyment and immersion, sparking greater user interest and participation. Gamification elements, such as challenges, rewards, and feedback, make the experience more interactive and engaging. Multi-sensory design, incorporating vision, hearing, smell, and touch, enriches the sensory experience, allowing users to fully appreciate the color, flavor, and texture of food, thus boosting both taste and psychological satisfaction.

Personalized design further enhances the experience by tailoring it to each user's preferences, health, and habits. Through smart analysis, the system offers Personalized user preference settings, ensuring each meal is a unique, tailored experience. This approach not only increases user engagement and satisfaction but also encourages the formation of healthy eating habits, improving long-term adherence to experience.

2.2 Basis of research

Theoretical support. Studies[9] show that music influences eating speed, experience, and mood. Slow music promotes mindful eating and enhances food enjoyment, while fast music can speed up eating and reduce food perception. Music also affects emotional reactions, relaxing tunes increase satisfaction, while unsuitable music may cause anxiety. The right music aids emotional regulation by reducing stress, improving stability, and enhancing focus, which is crucial for mindful eating.

UX Design theory emphasizes that user interaction with products or services goes beyond task completion; it also involves forming an emotional connection throughout the experience. The core idea is that user experience is shaped not only by functionality but also by the emotional resonance and sense of connection users develop. Factors like

ease of use, efficiency, satisfaction, and emotional response are crucial in shaping this experience. By optimizing these elements, designers can improve interaction quality, fostering stronger emotional attachment and loyalty. User experience design encourages designers to consider both the psychological and emotional needs of users, ensuring a seamless and satisfying interaction.

Technical Support. Mixed Reality (MR) blends the physical and digital worlds, enabling real-time interaction between real and virtual elements. Based on Milgram and Kishino's Virtuality Continuum Theory (1994), MR lies between Augmented Reality (AR), which adds virtual elements to the real world, and Augmented Virtuality (AV), which integrates real objects into the virtual world. Unlike VR, which fully immerses users in a digital environment, or AR, which overlays digital content onto the real world, MR allows seamless interaction with both physical and virtual objects. Devices like MR glasses or smartphones enhance immersion and create intuitive, photorealistic experiences, with applications in gaming, education, healthcare, design, and entertainment.

$$MR = AR + VR \quad (1)$$

This blending creates an immersive, interactive environment with real-time content manipulation, making MR an essential tool for many industries.

2.3 Research framework

This design combines mindful eating principles with multi-sensory interactions to create an immersive dining experience that promotes healthy habits and mindfulness. Recognizing that eating engages multiple senses, the system uses music and environmental cues to foster a positive atmosphere, helping alleviate negative emotions related to eating disorders or body anxiety. Through Augmented Reality (MR), users interact with virtual content in real-time, blending the virtual and physical worlds for enhanced immersion. The dual-platform system includes an app offering healthy eating tips, personalized plans, and progress tracking, while the MR platform enhances the mindful eating experience with virtual-real integration, food feedback, and generative art. This approach promotes healthier relationships with food, improves nutrition and mental health, and supports better eating habits and overall well-being. The system design is shown in Figure 1.



Fig 1. System design framework

2.4 Product mechanism

The Gourmet Flow system integrates Mixed Reality (MR) and AI-driven generative art to improve long-term adherence to mindful eating therapy. Using MR glasses and a

mobile app, it creates a personalized, immersive dining experience with real-time virtual interaction and AI-generated content. The process starts with the app analyzing food through the phone's camera, which is sent to the MR terminal to generate an adaptive virtual environment. Users can select and interact with their personalized dining setting, enhanced by interactive music visualization and scent-triggered feedback. This generates unique digital art combinations, fostering a multi-sensory, co-creative experience that improves both physical and mental satisfaction. The product mechanism and user behavior are illustrated in Table 1 and Figure 2.

Table 1.Product mechanism

Stage	Before eating	During eating smell	Eating taste	After eating
User behavior	Dishes, MR Virtual environment	Smell the real thing and choose	Taste and choose	View
Physical display	Set personal preferences	Choose the appropriate scent	Generate products based on taste	Record eating experience data.
stage	Before eating	Eating -Smell	Eating - Taste	After eating
Interactive Line	Pop up multiple scene selections :			
Front end behavior	Environment [Dawn Courtyard] Coastal Land Cloudspace Forest Stream	AI asks users what they smell : [Strong Aroma] [delicate fragrance] Sweet fragrance Sour Aroma spicy fragrance Frankincense	AI asks users about the taste they have tasted : sour [sweet] bitter Spicy Saltiness	Pop up situation record chart
Visible line of sight				
Background behavior	Prepare relevant data and resources based on user preferences and settings Different environments correspond to different music styles and tones	Corresponding instrument options Classical Symphony: Cello, Flute, Violin, French Horn, Oboe, Clarinet Folk Music: Acoustic Guitar, Harmonica, Banjo, Mandolin, Tambourine, Fiddle Pop Music: Piano, Guitar, Electronic Keyboard, Synthesizer, Electronic Drum, Electric Piano Rock: Drum, Electric Guitar, Bass, Amplifier, Electric Bass, Tom-tom Light Music: Hollow Drum, Celesta, Wind Chimes, Ukulele, Glockenspiel, Mando Piano		
		Corresponding color graphics Light Green + Butterfly Shape [Pink + Flower Shape] Purple + Hexagon Shape Orange + Four-Petaled Flower Blue + Octagon Shape		
		Analyze eating habits		

Some scenes and instrument melodies can only be unlocked after specific tasks are completed and certain conditions are met; users are required to respond to their selection operations within the specified time.

Fig.2 Schematic diagram of the product mechanism

3 Experimental results and analysis

3.1 Environment Setup

This subsection primarily outlines the modeling process of the experiment. Based on user preference research, it has been found that most people favor dreamy, aesthetic, and tranquil environments. Therefore, in scene design, light, low-saturation colors are chosen as the primary palette, with a harmonious tone that is gentle on the eyes and does not distract users from their meal. The design incorporates a surreal, ethereal atmosphere, infused with the vitality of a beautiful morning and the elegance of traditional Chinese courtyard architecture. These elements work together to help users achieve a state of calm and relaxation while eating. During the survey, we integrated principles from dietary psychology and environmental design to create multiple MR

scenarios aimed at enhancing user enjoyment during the mindful eating experience. By designing these scenarios, we strive to help users better focus on and savor their food, ultimately improving their overall dining experience while fostering healthier eating habits and psychological well-being. These MR scenarios not only increase immersion but also allow users to achieve more positive eating outcomes in a pleasant and relaxing atmosphere through interactive and sensory feedback from the virtual environment. The specific details are shown in Figure 3.

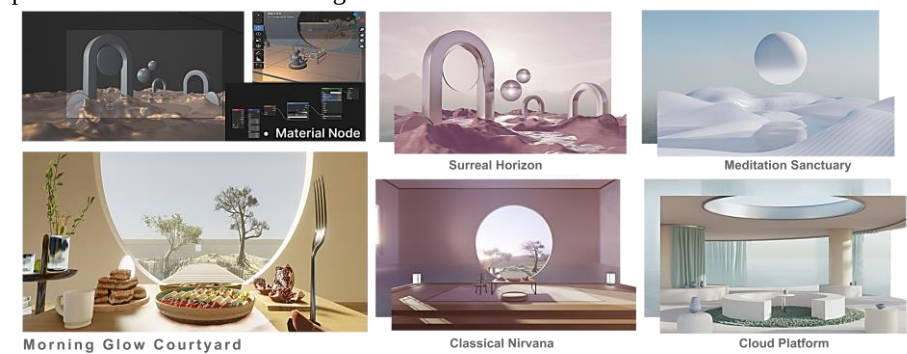


Fig. 3 Construction of MR Virtual Eating Environment

3.2 MR Interface Design

Apple’s MR interface design guidelines recommend keeping interface elements within 60% of the screen’s central area. This approach not only aligns with aesthetic principles but also enhances the ergonomic experience. In other words, interface content should remain within this 60% zone to ensure the design is both comfortable and functional.

MR systems should replicate real-world interactions to make the experience more intuitive and natural. Users should be able to interact with virtual elements using gestures, gaze, and voice, without relying on complex hardware or manipulation techniques. The interface design for MR systems prioritizes immersion, simplicity, and efficiency, blending real and virtual worlds seamlessly. It should minimize information overload, allowing users to easily identify and interact with necessary elements. The design should be clear and unobtrusive, with virtual elements integrating naturally into the real environment to maintain focus. The specific interface design is shown in Figure 4 and Table 2.

Table 2 The key interface design principles

Design Principle	Description
Centralized Interface Elements	Keep elements within 60% of the screen’s central area for better ergonomics and aesthetics.
Natural Interaction	Enable gestures, gaze, and voice interactions, avoiding complex hardware or manipulation.
Immersion & Simplicity	Prioritize a seamless blend of real and virtual worlds, focusing on simplicity and efficiency.
Unobtrusive Design	Ensure virtual elements integrate smoothly into the real environment to maintain user focus.



Fig. 4 The interface of MR

3.3 Artwork generation

The final art product is created using TouchDesigner, which transforms AI-generated music (in predefined styles) into dynamic visual art. The system integrates user feedback—gathered during the mindful eating experiences such as taste and smell, to adjust music parameters in real-time. This personalized music data is then sent to TouchDesigner, enabling real-time visualization of the auditory information. The closed-loop system ensures that distinct food perceptions trigger unique musical outputs, which generate customized visual artworks through TD's generative algorithms. This approach offers users a deeply personalized generative art experience, enhancing their mindful eating journey and promoting psychological well-being. The final effect is shown in Figure. 5

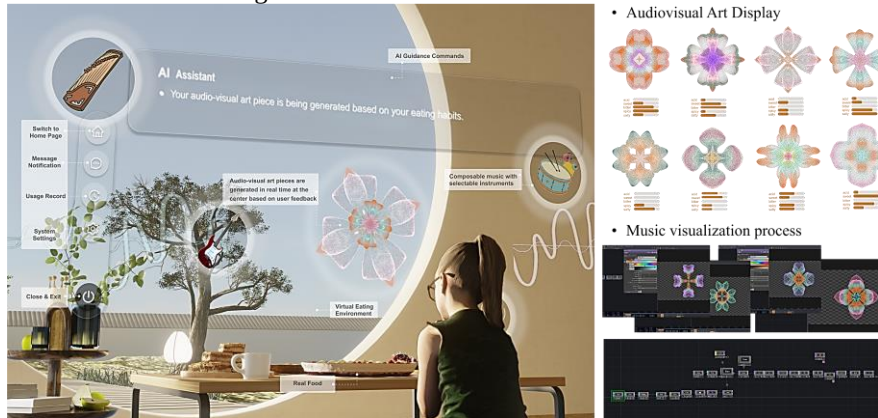


Fig.5 Final rendering

4 CONCLUSION

In conclusion, the proposed system, combining Mixed Reality (MR) and AI-driven generative art, offers an innovative solution to improving long-term adherence to

mindful eating. By providing personalized, immersive, and multi-sensory experiences, it addresses key limitations of traditional therapy, such as low engagement and lack of personalization. The Gourmet Flow system, with MR glasses, a mobile app, and real-time food data analysis, creates an adaptive, co-creative environment that enhances user interaction with virtual content. Interactive music and olfactory feedback provide a sensory-rich experience, promoting mindful eating, well-being, and psychological satisfaction.

Looking forward, future research could explore the system's scalability across diverse populations and cultures, refine AI algorithms for deeper personalization, and investigate its long-term effects on eating disorders and weight management. Expanding multi-sensory engagement through tactile feedback or taste simulation could further enrich the mindful eating experience.

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